



Air University, Islamabad

**Software Requirement Specification
(SRS DOCUMENT)**

for

Project Title

**Skill Up – Smart Freelancing Platform for Pakistan
Version 1.0**

By

Jawaad Mustafa 221796

Mehboob Ali 221824

Muhammad Furqan Saeed 221860

Supervisor

Mr Sharif Hussain

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Supervised by
Supervisor's Name

Signature_____

Table of Contents

1. Introduction	1
1.1 Purpose	1
1.2 Scope	1
1.3 Modules	2
1.4 Overview	3
2. Overall Description	3
2.1 Product Perspective	4
2.2 User classes and characteristics	5
2.3 Operating Environment	5
2.4 Design and Implementation Constraints	5
3. Requirement Identifying Technique	6
3.1 Use Cases	6
4. Functional Requirements	14
5. Non-Functional Requirements.....	20
5.1 Performance.....	20
5.2 Reliability	21
5.3 Security.....	21
5.4 Usability	21
5.5 Scalability	22
5.6 Maintainability	22
5.7 Portability	22
5.8 Compatibility	23
6. External Interface Requirements	23
6.1 User Interfaces Requirements	23
6.2 Software Interfaces	29
6.3 Hardware Interfaces	29
6.4 Communication Interfaces	29
7. References	30

1. Introduction

The introduction of the Software Requirements Specification (SRS) should provide an overview of the entire SRS. It should include the purpose, scope, modules and overview of the SRS.

1.1 Purpose

This Software Requirements Specification (SRS) document defines the requirements for SkillUp – Ai Base Smart Freelancing Platform for Pakistan. It serves as a formal agreement between the development team, supervisor, and all stakeholders. The primary audience of this document includes:

- Developers – for understanding the system’s functional and technical requirements.
- Testers – for verifying that the delivered product meets the documented requirements.
- Clients, Freelancers, and Admins – as primary end users who interact with the system.
- UI/UX Designers – for designing user interfaces aligned with user expectations.

This document outlines all functional and non-functional requirements, interface requirements, system constraints, and describes the system architecture to be implemented.

1.2 Scope

SkillUp is a web-based freelancing platform designed specifically to serve freelancers and clients in Pakistan. The system addresses major challenges in the local freelancing industry, including high commission fees, lack of local payment gateways, oversaturation on global platforms, and the absence of professionalism on informal platforms like Facebook and WhatsApp. SkillUp incorporates features such as:

- AI-powered job matching
- Secure Escrow-based payment handling
- Integration with JazzCash, Easypaisa, and bank transfers
- Freelancer profile creation, portfolio uploads, and skill testing
- Project posting, bidding, milestone tracking, and reviews
- Proposal and job description generation assistance via AI
- Comprehensive admin dashboard for dispute resolution and platform management

The system will initially support only Pakistani users and payment systems. It aims to provide a trustworthy, localized, and accessible freelancing environment with low commission rates (5%) and AI-enhanced user experiences

1.3 Modules

The system is divided into the following main functional modules, each serving a distinct role in delivering platform functionality:

- **Module 1: User Authentication**

FE-1: User registration and login for Freelancers and Clients.

FE-2: CNIC verification for authenticity.

FE-3: Email verification and password recovery system.

FE-4: Role-based access control (Freelancer, Client, Admin).

- **Module 2: Freelancer Dashboard**

FE-1: Create and manage personal profile.

FE-2: Upload and manage portfolio.

FE-3: Take skill tests and earn certifications.

FE-4: View and apply for jobs.

FE-5: Track bids, ongoing jobs, earnings, and reviews.

FE-6: Manage milestones and communication with clients.

- **Module 3: Client Dashboard**

FE-1: Post jobs/projects with details.

FE-2: View freelancer profiles and proposals.

FE-3: Shortlist and hire freelancers.

FE-4: Manage ongoing projects and payments.

FE-5: Rate and review freelancers.

- **Module 4: AI-Based Job Matching**

FE-1: Recommend jobs to freelancers based on profile.

FE-2: Recommend freelancers to clients based on job descriptions.

FE-3: Continuously improve matching algorithm with feedback.

FE-4: AI-assisted proposal description generation.

- **Module 5: Payment System**

FE-1: Escrow-based payment system for secure transactions.

FE-2: Milestone creation and payment release.

FE-3: Integration with JazzCash, Easypaisa, and Bank Transfers.

FE-4: Transaction history and withdrawal management.

FE-5: Admin support for dispute resolution.

- **Module 6: Project Posting & Browsing**

- FE-1:** Clients can post job listings with specific requirements.
- FE-2:** Freelancers can browse and filter job listings.
- FE-3:** Save jobs and receive notifications.
- FE-4:** Display project details, client ratings, and bid count.

- **Module 7: Bidding & Proposal Handling**

- FE-1:** Freelancers can submit proposals with bids.
- FE-2:** Clients can view and compare proposals.
- FE-3:** Messaging system for client-freelancer communication.
- FE-4:** Notification alerts for proposal acceptance/rejection.

- **Module 8: Admin Panel**

- FE-1:** Admin dashboard for platform stats and user activity.
- FE-2:** Manage user accounts (verify, suspend, delete).
- FE-3:** Review job postings and proposals for violations.
- FE-4:** Handle disputes and refund requests.
- FE-5:** Generate reports and manage announcements.

1.4 Overview

The remainder of this document is structured as follows:

- **Section 2** provides a high-level product description, including system context, user classifications, operating environment, and constraints.
- **Section 3** outlines the techniques used to derive and capture system requirements.
- **Section 4** describes the functional requirements organized by module.
- **Section 5** outlines non-functional requirements such as performance, reliability, usability, and security.
- **Section 6** details the system's external interface requirements, including user interface, software, hardware, and communication interfaces.
- **Section 7** lists all references and citations used in the project and documentation.

2. Overall Description

This section presents a high-level overview of the product and the environment in which it will be used, the anticipated users, and known constraints, assumptions, and dependencies.

2.1 Product Perspective

SkillUp is a newly developed web-based platform tailored for the Pakistani freelancing market. It is an original product built from scratch with no dependency on existing platforms. SkillUp is not an extension of any previously developed system. It is intended to serve as a localized alternative to international freelancing marketplaces such as Upwork and Fiverr by addressing the specific needs of Pakistani users. SkillUp introduces AI-powered automation, secure local payment handling, freelancer skill validation, and a structured, verified platform for project-based work.

The system will interact with external services via APIs, including:

- OpenAI API for AI-based job matching and proposal generation.
- JazzCash, Easypaisa, and local bank APIs for payment processing.
- NADRA for CNIC verification

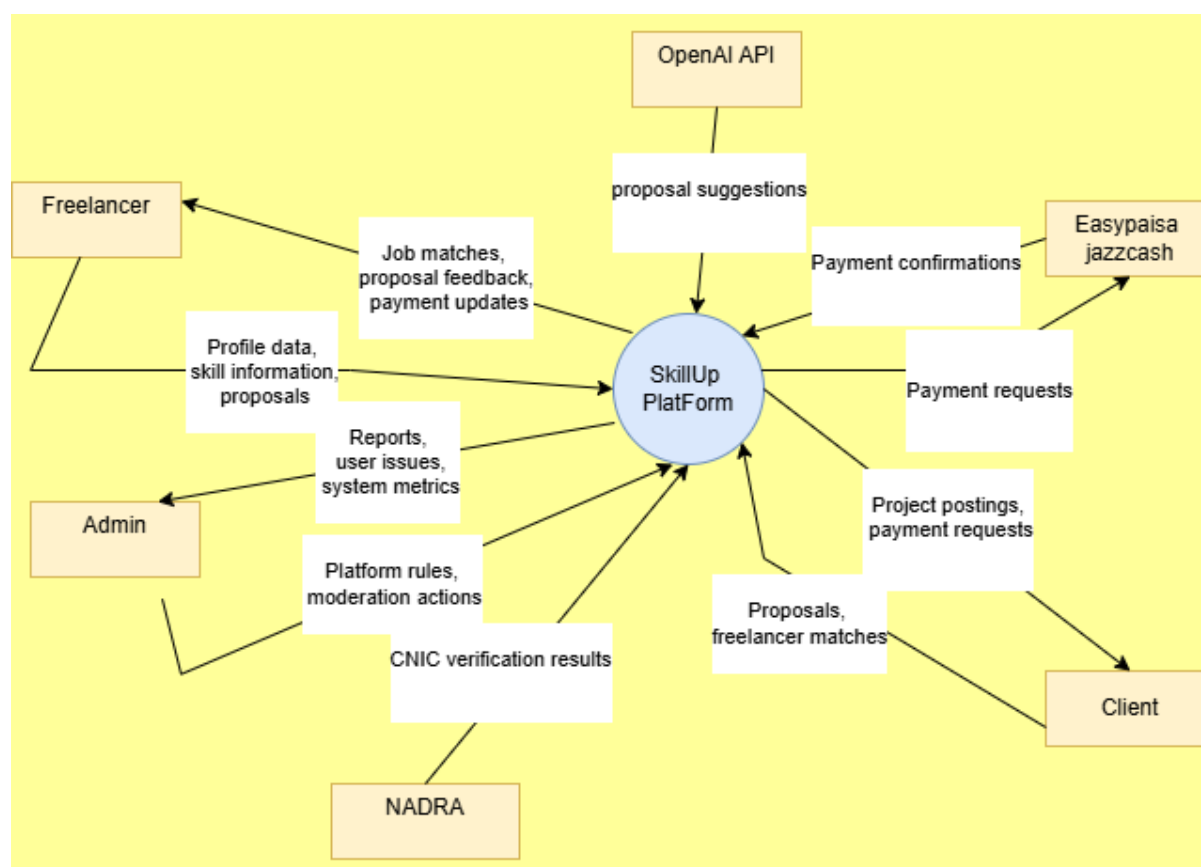


Fig 1.0: Context Diagram of Skill-up Platform

2.2 User classes and characteristics

User Class	Description
Freelancers	Users who register to find, apply for, and complete freelance jobs. May include students, graduates, and professionals. Will be able to build portfolios, take skill tests, receive certifications, submit proposals, and receive payments. Users will need to verify their identity and take initial onboarding assessments.
Clients	Businesses or individuals looking to hire freelancers for various digital services. Can post projects, review proposals, hire talent, release payments, and leave reviews. May range from small business owners to startups and digital marketing firms within Pakistan.
Admin	Platform owners or representatives responsible for system oversight. Manage user accounts, verify identities, handle disputes, and monitor system activity. Will have access to a dedicated admin dashboard with platform analytics and controls

2.3 Operating Environment

SkillUp will be a web-based platform accessible via modern browsers on desktop and mobile devices. The application will be deployed on a cloud-based infrastructure (such as AWS or DigitalOcean) and will run within a containerized environment using Docker.

OE-1: The system shall operate correctly on the latest versions of Google Chrome, Mozilla Firefox, and Microsoft Edge.

OE-2: The backend server will run on a Linux-based environment (e.g., Ubuntu 22.04 LTS).

OE-3: The platform shall be hosted on a cloud provider that supports HTTPS, SSL/TLS, and automatic scaling.

OE-4: The database will reside on a managed MongoDB instance with replication enabled for reliability.

OE-5: Freelancers and clients will typically access the platform from within Pakistan, requiring support for local networks and occasional high latency.

2.4 Design and Implementation Constraints

CON-1: The platform must use the MERN stack (MongoDB, Express.js, React.js, Node.js).

CON-2: The system must integrate with OpenAI's API for AI-powered features such as job matching and proposal writing.

CON-3: Payment systems must use local gateways such as JazzCash, Easypaisa, and local bank APIs

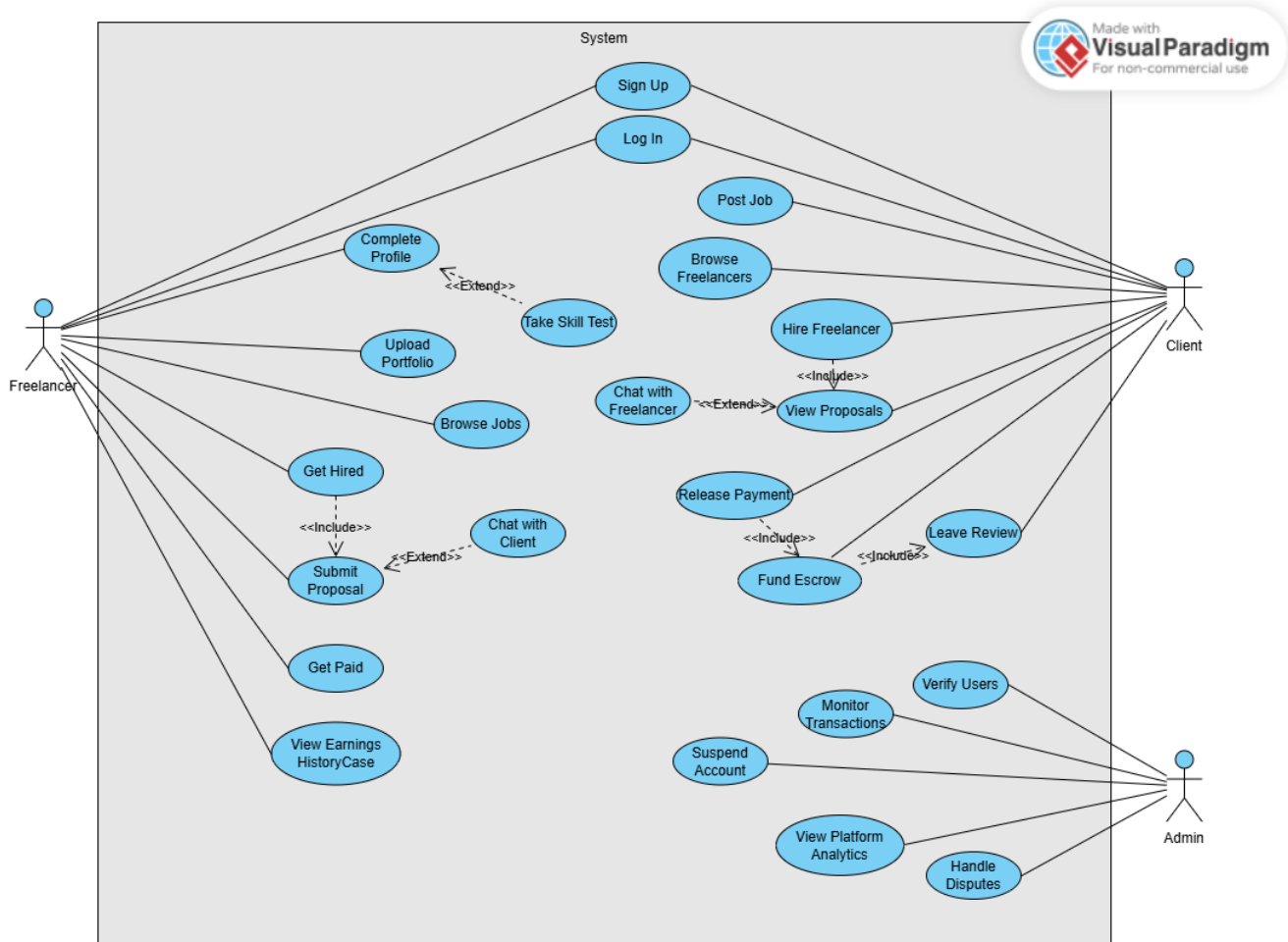
CON-4: Docker containers must be used for deployment to ensure cross-environment compatibility and ease of CI/CD.

CON-5: User authentication must support CNIC verification, email verification, and secure password management.

CON-6: Mobile app support will not be available in the MVP version; only a responsive web application will be delivered.

3. Requirement Identifying Technique

3.1 Use Cases



Use Case Table – UC-1: Freelancer Signup

Field	Description
Use Case ID	UC-1
Use Case Name	Freelancer Signup
Actors	Freelancer (Primary) Authentication Service (System Actor)
Description	This use case allows a new freelancer to create an account on SkillUp using a unique email and secure password.
Trigger	Freelancer clicks the “Sign Up” button on the homepage.
Preconditions	PRE-1: Freelancer must not already have an account with the same email. PRE-2: Internet connection is active.
Postconditions	POST-1: A verified freelancer account is created. POST-2: User is redirected to the dashboard upon successful signup and verification.
Normal Flow	1. Freelancer clicks “Sign Up”. 2. System displays registration form. 3. Freelancer enters required details. 4. Clicks Submit. 5. System validates inputs. 6. Sends verification email. 7. Freelancer clicks verification link. 8. Redirects to dashboard.
Alternative Flows	5.1: If email already exists Show “Email already registered” Return to Step 2 5.2: Weak password , Show message Return to Step 3
Exceptions	E1: Server/database is unreachable Show “Signup failed. Try again later.”
Business Rules	BR-1: Email must be unique. BR-2: Password must be encrypted before storing. BR-3: Email must be verified before login is allowed.
Assumptions	AS-1: User has access to a valid email account. AS-2: Authentication system is online and operational.

Use Case Table – UC-2: Client Signup

Field	Description
Use Case ID	UC-2
Use Case Name	Client Signup
Actors	Client (Primary), Authentication Service
Description	Allows a new client to register with their email and password to post jobs on the platform.
Trigger	Client clicks “Sign Up” from the homepage.

Preconditions	PRE-1: Client has not already registered with the same email. PRE-2: Internet is accessible.
Postconditions	POST-1: Client account is successfully stored in the system. POST-2: Client can now log in and post jobs.
Normal Flow	1. Client clicks “Sign Up”. 2. System shows form. 3. Client enters email and password. 4. Clicks submit. 5. System validates and stores info. 6. Email verification is sent.
Alternative Flows	4.1: Email already exists then show error and return to form 4.2: Password too weak then show alert and request update
Exceptions	E1: Backend/database is offline then show “Signup failed”
Business Rules	BR-1: Unique email required. BR-2: Password must be encrypted. BR-3: Client must accept Terms and Conditions before signup is allowed.
Assumptions	AS-1: Client has a valid email address. AS-2: Platform services (database, email) are online.

Use Case Table – UC-3: Login

Field	Description
Use Case ID	UC-3
Use Case Name	User Login
Actors	Freelancer, Client
Description	Allows users to securely log into their SkillUp account using email and password.
Trigger	User clicks the “Login” button.
Preconditions	PRE-1: User account is verified. PRE-2: Internet access is available.
Postconditions	POST-1: User is authenticated. POST-2: User is redirected to their respective dashboard.
Normal Flow	1. User clicks “Login”. 2. Enters email and password. 3. Clicks “Submit”. 4. System authenticates credentials. 5. Redirects user.
Alternative Flows	4.1: Invalid credentials then show “Incorrect email/password”
Exceptions	E1: Authentication server fails and display error message
Business Rules	BR-1: Password must be securely hashed.

	BR-2: Login should time out after 30 minutes of inactivity.
Assumptions	AS-1: User remembers valid login credentials. AS-2: Authentication system is working.

Use Case Table – UC-4: Post Job

Field	Description
Use Case ID	UC-4
Use Case Name	Post Job
Actors	Client (Primary)
Description	This use case allows a registered client to post a new freelance job with details like budget, description, and required skills.
Trigger	Client clicks on the “Post a Job” button in the dashboard.
Preconditions	PRE-1: Client is logged in. PRE-2: Client has verified their account.
Postconditions	POST-1: The job is listed and becomes visible to freelancers. POST-2: Notifications are sent to relevant freelancers.
Normal Flow	1. Client logs in. 2. Clicks “Post Job”. 3. Fills job form (title, description, budget, deadline, required skills). 4. Clicks Submit. 5. System validates and saves job listing. 6. Job appears on platform.
Alternative Flows	4.1: Required field missing then show validation error and return to form 4.2: Invalid budget format then show alert
Exceptions	E1: System crash during submission then show “Job post failed. Try again.”
Business Rules	BR-1: Budget must be in PKR. BR-2: Job title must be 10–100 characters. BR-3: Description must be a minimum of 100 characters.
Assumptions	AS-1: Client has all project information prepared (e.g., budget, scope). AS-2: Platform backend is responsive.

Use Case Table – UC-5: Submit Proposal

Field	Description
Use Case ID	UC-5
Use Case Name	Submit Proposal
Actors	Freelancer (Primary)
Description	This use case allows freelancers to respond to a job post by submitting a formal proposal.
Trigger	Freelancer clicks the “Apply” button on a job listing.

Preconditions	PRE-1: Freelancer is logged in. PRE-2: Freelancer's profile is at least 60% complete. PRE-3: Freelancer has not exceeded weekly proposal limit.
Postconditions	POST-1: Proposal is stored and visible to the client. POST-2: Freelancer receives confirmation.
Normal Flow	1. Freelancer logs in. 2. Clicks on a job post. 3. Clicks "Apply". 4. Enters cover letter, price, delivery time, and uploads files. 5. Clicks Submit. 6. System stores the proposal and notifies client.
Alternative Flows	5.1: Proposal fields left empty the show "Please complete all fields" and return to Step 4 5.2: File upload fails then show upload error
Exceptions	E1: Proposal submission fails due to backend error then show "Unable to apply. Try again later."
Business Rules	BR-1: Freelancers can submit max 20 proposals/week. BR-2: Files must be under 10 MB. BR-3: AI-generated proposals must be editable.
Assumptions	AS-1: Freelancer has the necessary documents/portfolio ready. AS-2: Job post is still active.

Use Case Table – UC-6: Hire Freelancer

Field	Description
Use Case ID	UC-6
Use Case Name	Hire Freelancer
Actors	Client (Primary)
Description	This use case allows a client to hire a freelancer and define project milestones with associated escrow payments.
Trigger	Client clicks "Hire" on a selected proposal.
Preconditions	PRE-1: Client is logged in. PRE-2: Client has a verified payment method. PRE-3: Proposal is in "shortlisted" or "pending" state.
Postconditions	POST-1: Contract is created. POST-2: Funds are locked in escrow.
Normal Flow	1. Client reviews proposals. 2. Clicks "Hire". 3. System prompts for milestone setup. 4. Client defines at least one milestone. 5. System processes escrow deposit.

	6. Confirmation is sent to freelancer.
Alternative Flows	3.1: Escrow deposit fails the show “Payment failed” and option to retry or cancel 4.1: Client cancels hiring then return to proposal list
Exceptions	E1: Backend fails during hire confirmation then show “Unable to complete contract. Try again.”
Business Rules	BR-1: At least one milestone is required. BR-2: Escrow must be funded before project begins. BR-3: Hire action is final (editable only via admin).
Assumptions	AS-1: Freelancer is available. AS-2: Escrow and milestone modules are active and functional.

Use Case Table – UC-7: CNIC Verification

Field	Description
Use Case ID	UC-7
Use Case Name	CNIC Verification
Actors	Freelancer (Primary), Verification System (System Actor)
Description	This use case allows freelancers to verify their identity by submitting a valid CNIC, which is then cross-checked via a verification service.
Trigger	Freelancer attempts to complete profile setup.
Preconditions	PRE-1: Freelancer is logged in. PRE-2: Profile completion is at or below 80%.
Postconditions	POST-1: Freelancer account status is marked as “Verified”. POST-2: Verified badge is shown on public profile.
Normal Flow	1. Freelancer navigates to CNIC verification. 2. Enters CNIC number and uploads scanned document. 3. System sends request to verification API. 4. Upon success, account is marked verified. 5. Notification sent to user.
Alternative Flows	3.1: CNIC rejected then system shows “Invalid CNIC, please recheck” and return to Step 2
Exceptions	E1: Verification system offline then show “Service temporarily unavailable. Try again later.”
Business Rules	BR-1: One CNIC can be used for only one account. BR-2: CNIC must be 13 digits and match uploaded image.
Assumptions	AS-1: Freelancer has a valid Pakistani CNIC. AS-2: Third-party or internal verification system is online.

Use Case Table – UC-8: Take Skill Test

Field	Description
Use Case ID	UC-8
Use Case Name	Take Skill Test
Actors	Freelancer
Description	This use case allows freelancers to take skill-based quizzes for various services (e.g., design, writing), and earn certifications upon passing.
Trigger	Freelancer clicks “Take Skill Test” from their dashboard.
Preconditions	PRE-1: Freelancer is logged in. PRE-2: Freelancer has a completed profile.
Postconditions	POST-1: Score and result are stored. POST-2: If passed, badge is added to profile.
Normal Flow	1. Clicks “Take Skill Test”. 2. Selects a skill category. 3. System loads test. 4. Freelancer answers MCQs. 5. Submits test. 6. System scores and displays results.
Alternative Flows	5.1: Time runs out and system auto-submits test
Exceptions	E1: Test database not loading then show error and disable test temporarily
Business Rules	BR-1: Test duration = 20 minutes. BR-2: Minimum passing score = 70%. BR-3: Badge shown only after passing.
Assumptions	AS-1: Freelancer is allowed to take up to 3 attempts per skill. AS-2: Questions are randomly selected from a test bank.

Use Case Table – UC-9: Escrow Payment & Release

Field	Description
Use Case ID	UC-9
Use Case Name	Escrow Payment & Release
Actors	Client, Freelancer, Payment Gateway (System Actor)
Description	This use case manages the deposit of funds into escrow by clients, and release upon milestone completion with freelancer approval.
Trigger	A client funds a milestone and marks it as complete.
Preconditions	PRE-1: Milestone is defined and agreed by both parties. PRE-2: Client has a verified payment method.
Postconditions	POST-1: Payment is released from escrow to freelancer’s wallet. POST-2: Transaction is logged and visible to both users.
Normal Flow	1. Client funds escrow for milestone. 2. Freelancer submits deliverable.

	3. Client marks milestone as complete. 4. Freelancer approves release. 5. System releases payment.
Alternative Flows	3.1: Client fails to confirm delivery and reminder sent every 24 hours
Exceptions	E1: Escrow gateway fails and show “Unable to release payment now.”
Business Rules	BR-1: Funds remain locked until both parties approve. BR-2: Admin can override in dispute cases.
Assumptions	AS-1: Escrow service is functioning. AS-2: Freelancer has delivered the agreed work.

Use Case Table – UC-10: Messaging Between Client and Freelancer

Field	Description
Use Case ID	UC-10
Use Case Name	Messaging Between Client and Freelancer
Actors	Freelancer, Client
Description	This use case allows real-time communication between clients and freelancers through a built-in chat system.
Trigger	A proposal is submitted, or a job is hired.
Preconditions	PRE-1: Both users are logged in. PRE-2: A valid proposal or contract exists between them.
Postconditions	POST-1: Messages are logged and accessible. POST-2: Notifications are triggered on new messages.
Normal Flow	1. User opens a job chat. 2. Writes and sends message. 3. Message is delivered and stored. 4. Receiver gets real-time notification.
Alternative Flows	3.1: Network delay and retry message delivery
Exceptions	E1: Messaging backend fails and show “Message failed. Try again later.”
Business Rules	BR-1: Users may not share phone numbers or external links. BR-2: Admin may monitor chats for T&C violations.
Assumptions	AS-1: Messaging module is active. AS-2: Users comply with community guidelines.

Use Case Table – UC-11: Admin Dispute Resolution

Field	Description
Use Case ID	UC-11
Use Case Name	Admin Dispute Resolution

Actors	Admin (Primary), Freelancer, Client
Description	Admin can review disputes between freelancers and clients, and take actions such as approving refunds, warnings, or account suspension.
Trigger	Either party files a dispute, or flags a transaction.
Preconditions	PRE-1: Dispute is submitted. PRE-2: Admin has platform access.
Postconditions	POST-1: Case is resolved. POST-2: Outcome (e.g., refund issued, warning applied) is logged.
Normal Flow	1. Admin opens dispute case. 2. Reviews timeline, messages, transactions. 3. Contacts both parties (if needed). 4. Chooses outcome. 5. System applies and records action.
Alternative Flows	3.1: One party unresponsive then Admin rules based on available evidence
Exceptions	E1: Admin dashboard down and resolution is postponed
Business Rules	BR-1: Disputes must be resolved within 72 hours. BR-2: Admin must remain impartial and follow platform policy.
Assumptions	AS-1: Admin is trained to evaluate cases. AS-2: Platform keeps full logs of messages and transactions.

4. Functional Requirements

Table 4.1: Description of FR-1

Identifier	FR-1
Title	User Registration and Login
Requirement	The system shall allow users (Freelancers and Clients) to register, log in, and log out securely using their email and password.
Source	Project Proposal – Module 7.1
Rationale	To establish authenticated access and personalized use of the system.
Business Rule (if required)	Email must be unique. Password must meet minimum strength requirements (at least 8 characters, 1 uppercase, 1 symbol).
Dependencies	None
Priority	High

Table 4.2: Description of FR-2

Identifier	FR-2
Title	CNIC Verification
Requirement	The system shall require Freelancers to submit a valid CNIC number during registration, which will be verified using a third-party API.

Source	Project Proposal – Module 7.1
Rationale	To ensure legitimacy of users and prevent fake or duplicate accounts.
Business Rule (if required)	A verified CNIC cannot be used to create more than one account.
Dependencies	FR-1
Priority	High

Table 4.3: Description of FR-3

Identifier	FR-3
Title	Email Verification and Password Recovery
Requirement	The system shall send a verification link to the user's email upon registration and allow password reset through a secure link.
Source	Module 7.1
Rationale	To ensure valid email ownership and allow recovery from forgotten credentials.
Business Rule (if required)	Password reset links shall expire in 30 minutes.
Dependencies	FR-1
Priority	High

Table 4.4: Description of FR-4

Identifier	FR-4
Title	Freelancer Profile Management
Requirement	Freelancers shall be able to create and update their profile, including name, location, profile image, education, bio, and list of skills.
Source	Module 7.2
Rationale	To enable clients to assess freelancer qualifications and for AI job-matching to operate effectively.
Business Rule (if required)	A minimum profile completeness of 60% is required before bidding on jobs.
Dependencies	FR-1, FR-2
Priority	High

Table 4.5: Description of FR-5

Identifier	FR-5
Title	Portfolio Upload
Requirement	The system shall allow freelancers to upload sample work as images, files, or links to build a portfolio.
Source	Module 7.2
Rationale	To help freelancers showcase their capabilities and past experience to attract clients.

Business Rule (if required)	Uploaded files must be under 10 MB and scanned for malware.
Dependencies	FR-4
Priority	Medium

Table 4.6: Description of FR-6

Identifier	FR-6
Title	Skill Tests and Certification
Requirement	Freelancers shall be able to attempt online skill tests for specific categories. Upon passing, a certificate shall be displayed on their profile.
Source	Module 7.2
Rationale	To enable a verifiable trust mechanism that boosts credibility and improves hiring chances.
Business Rule (if required)	Users can attempt a test a maximum of 3 times. Tests are automatically scored.
Dependencies	FR-4
Priority	Medium

Table 4.7: Description of FR-7

Identifier	FR-7
Title	Apply for Jobs and Track Proposals
Requirement	Freelancers shall browse jobs and submit proposals. They shall also view the status of each proposal (submitted, shortlisted, accepted, rejected).
Source	Module 7.2
Rationale	To allow freelancers to actively seek opportunities and monitor their job applications.
Business Rule (if required)	Freelancers may only apply to a maximum of 20 jobs per week (adjustable via admin panel).
Dependencies	FR-4, FR-11, FR-13
Priority	High

Table 4.8: Description of FR-8

Identifier	FR-8
Title	Post Job/Project
Requirement	Clients shall be able to post new jobs or project listings including job title, description, budget, required skills, deadline, and attachments.
Source	Module 7.3
Rationale	To allow clients to formally offer opportunities for freelancers to apply.
Business Rule (if required)	Budget must be specified in PKR. Required skills must be selected from a predefined list.
Dependencies	FR-1
Priority	High

Table 4.9: Description of FR-9

Identifier	FR-9
Title	Review Freelancer Proposals
Requirement	Clients shall be able to view all received proposals for a posted job, compare them, and shortlist or reject candidates.
Source	Module 7.3
Rationale	To help clients evaluate candidates and proceed with hiring decisions.
Business Rule (if required)	Clients must respond to proposals within 7 days to maintain platform trust.
Dependencies	FR-8, FR-18
Priority	High

Table 4.10: Description of FR-10

Identifier	FR-10
Title	Hire Freelancers and Manage Jobs
Requirement	Clients shall be able to formally hire a freelancer, initiate a contract, set milestones, and monitor job progress.
Source	Modules 7.3, 7.5
Rationale	To formalize the working relationship and provide tools for transparent project management.
Business Rule (if required)	A project must have at least one milestone and one assigned freelancer.
Dependencies	FR-9, FR-14
Priority	High

Table 4.11: Description of FR-11

Identifier	FR-11
Title	Job Recommendations for Freelancers
Requirement	The system shall analyze a freelancer's profile, skills, and history using AI and suggest relevant job postings.
Source	Module 7.4
Rationale	To save time and increase relevance by automatically showing jobs that fit each freelancer.
Business Rule (if required)	Recommendations should refresh every 12 hours or when the profile is updated.
Dependencies	FR-4, FR-7
Priority	High

Table 4.12: Description of FR-12

Identifier	FR-12
Title	Freelancer Recommendations for Clients

Requirement	When clients post a job, the system shall use AI to automatically recommend the most suitable freelancers for that job.
Source	Module 7.4
Rationale	To speed up hiring and improve the quality of freelancer selection.
Business Rule (if required)	At least three freelancers should be suggested based on match score.
Dependencies	FR-8
Priority	High

Table 4.13: Description of FR-13

Identifier	FR-13
Title	Proposal Generation via AI
Requirement	The system shall generate smart proposals using AI based on job description, freelancer profile, and previous successful proposal patterns.
Source	Module 7.4
Rationale	To help freelancers write better proposals and improve their chances of selection.
Business Rule (if required)	AI-generated content must be editable before submission.
Dependencies	OpenAI API, FR-7
Priority	Medium

Table 4.14: Description of FR-14

Identifier	FR-14
Title	Escrow-based Transaction System
Requirement	The system shall hold client funds in a secure escrow account upon hiring a freelancer and release the funds to the freelancer upon mutual milestone approval.
Source	Module 7.5
Rationale	To ensure payment security and build trust between clients and freelancers.
Business Rule (if required)	Funds cannot be released without approval from both client and freelancer; admin may intervene in case of disputes.
Dependencies	FR-10
Priority	High

Table 4.15: Description of FR-15

Identifier	FR-15
Title	Milestone Management
Requirement	Clients and freelancers shall be able to define, edit, and approve project milestones, each associated with a budget and deadline.
Source	Module 7.5

Rationale	To manage large projects in smaller deliverable phases and enable partial payments.
Business Rule (if required)	At least one milestone must be defined before starting a job. Milestones must be approved before payment release.
Dependencies	FR-10, FR-14
Priority	Medium

Table 4.16: Description of FR-16

Identifier	FR-16
Title	Local Payment Integration
Requirement	The system shall integrate APIs of JazzCash, Easypaisa, and local banks to process freelancer withdrawals and client deposits.
Source	Module 7.5
Rationale	To support faster and locally compatible payment systems for Pakistani users.
Business Rule (if required)	Users must verify payment details (e.g., account number or mobile wallet) before initiating a transaction.
Dependencies	FR-14
Priority	High

Table 4.17: Description of FR-17

Identifier	FR-17
Title	Transaction History
Requirement	The system shall maintain detailed logs of all transactions including payment amounts, timestamps, involved parties, and status (pending, completed, failed).
Source	Module 7.5
Rationale	To provide transparency and enable users to track their financial history.
Business Rule (if required)	Transaction records must be immutable after confirmation.
Dependencies	FR-14, FR-16
Priority	Medium

Table 4.18: Description of FR-18

Identifier	FR-18
Title	Submit and Manage Proposals
Requirement	Freelancers shall be able to submit proposals for jobs including cover letters, pricing, delivery time, and optional file attachments.
Source	Module 7.7
Rationale	To allow freelancers to formally apply for job opportunities with relevant details.

Business Rule (if required)	Each freelancer can send up to 20 active proposals per week; expired proposals are archived automatically.
Dependencies	FR-7, FR-13
Priority	High

Table 4.19: Description of FR-19

Identifier	FR-19
Title	Client-Freelancer Messaging
Requirement	The system shall provide a secure in-platform messaging system for freelancers and clients to communicate before and during a project.
Source	Module 7.7
Rationale	To facilitate collaboration, question clarification, and project alignment.
Business Rule (if required)	Users must not exchange personal contact details; messages are monitored to prevent terms violation.
Dependencies	FR-18
Priority	High

Table 4.20: Description of FR-20

Identifier	FR-20
Title	Admin Dashboard and User Management
Requirement	The system shall provide an admin dashboard where authorized users can view platform stats, manage user accounts, and resolve disputes.
Source	Module 7.8
Rationale	To ensure platform moderation, compliance, dispute resolution, and integrity of operations.
Business Rule (if required)	Only verified admin accounts can access the dashboard. Admins can suspend or reinstate accounts based on investigation.
Dependencies	All major modules
Priority	High

5. Non-Functional Requirements

5.1 Performance

NFR-P1: 95% of web pages shall load completely within 3 seconds on a 10 Mbps or faster internet connection.

NFR-P2: Job matching suggestions through AI for common freelancer or job queries shall be returned in less than 2 seconds.

NFR-P3: The system shall support up to 10,000 simultaneous users, with performance

degradation at peak hours.

NFR-P4: The Escrow and Payment module shall process internal transactions within 1 second (excluding third-party API response time).

5.2 Reliability

NFR-R1: The system shall maintain 99.5% uptime availability, excluding scheduled maintenance windows.

NFR-R2: Using auto-restart services in a deployment environment will ensure the system recovers from unexpected crashes or service interruptions within a maximum of 30 seconds.

NFR-R3: All important data (user profiles, transactions, bids, messages) shall be backed up every 6 hours and stored on a secure, encrypted cloud location.

NFR-R4: The system shall gracefully handle failed API calls (such as payment gateway timeouts) by displaying a retry message and logging the failure event.

5.3 Security

NFR-S1: All data transmission shall be encrypted using SSL/TLS (HTTPS).

NFR-S2: Passwords shall be stored using salted hashing (such as bcrypt) and will never be stored in plain text.

NFR-S3: User roles (freelancer, client, admin) shall be enforced at all levels through Role-Based Access Control (RBAC).

NFR-S4: The system shall automatically log out inactive users after 30 minutes of inactivity.

NFR-S5: CNIC numbers and other personal identification data shall be encrypted and stored only after user want.

NFR-S6: All financial transactions shall be logged with audit trails including timestamps, transaction IDs, and IP addresses.

NFR-S7: Admin dashboard shall require multi-factor authentication (MFA).

5.4 Usability

NFR-U1: The platform shall support both Urdu and English language interfaces to expand its reach to a wider Pakistani user base.

NFR-U2: The user interface shall be responsive and compatible with all major screen sizes, including desktops, tablets, and smartphones.

NFR-U3: New users shall be able to register, complete profile setup, and browse jobs or post

projects within 10 minutes.

NFR-U4: All features shall be accessible within a maximum of three user interactions (clicks/taps) from dashboard.

NFR-U5: The interface shall follow modern usability standards including keyboard shortcuts, clean layouts, consistent button styling, and tooltips where necessary.

5.5 Scalability

NFR-Sc1: The system shall support horizontal scaling and handle a 3x increase in users without the need for architectural redesign.

NFR-Sc2: The backend shall be modular to allow distribution across multiple servers or containers.

NFR-Sc3: The database shall be optimized to scale with data growth and concurrent access demands.

5.6 Maintainability

NFR-M1: The codebase shall follow industry standard naming conventions and be structured for modular development.

NFR-M2: Developer documentation shall be maintained for all API endpoints, modules, and deployment processes.

NFR-M3: The platform shall support bug tracking and version control through tools like GitHub.

NFR-M4: System logs shall be saved and organized to allow for error detection and system monitoring.

5.7 Portability

NFR-Pt1: The system shall be containerized (for example, via Docker) to allow easy deployment on various cloud platforms.

NFR-Pt2: The application components (frontend, backend, database) shall be independently deployable.

NFR-Pt3: The application shall support migration between cloud providers without loss of data or functionality.

5.8 Compatibility

NFR-C1: The platform shall work properly on the latest versions of Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge.

NFR-C2: The mobile version shall be compatible with Android 8+ and iOS 12+ devices.

NFR-C3: The system shall offer a simplified low-bandwidth mode to ensure use on slow internet connections.

6. External Interface Requirements

6.1 User Interfaces Requirements

The user interface (UI) of the SkillUp platform shall be designed with a focus on accessibility, responsiveness, modern aesthetics, and ease of use. It will follow a structured layout and standardized visual language to provide a consistent and intuitive user experience for both freelancers and clients.

6.1.1 GUI Standards and Style Guide

- The interface will follow modern web GUI standards using the **Tailwind CSS** utility-first framework.
- A consistent **component-based design system** shall be implemented for reusable UI elements (buttons, modals, forms).
- Icons will be sourced from the **Lucide React** icon library to ensure visual consistency.
- The system shall follow **WCAG 2.1 AA** accessibility guidelines for inclusive design.

6.1.2 Fonts, Colors, and Icons

- **Fonts:**
 - **Primary (English):** Inter, system-ui, sans-serif
 - **Secondary (Urdu):** Noto Nastaliq Urdu
 - Font sizes:
 - Body text: 12px (regular)
 - Headings: H1 – 24px, H2 – 20px, H3 – 18px
 - Small text: 12px for tooltips and metadata
- **Color Scheme:**
 - Primary: Indigo #4f46e5
 - Success: Emerald #10b981
 - Warning: Amber #f59e0b
 - Error: Rose #ef4444
 - Background: Light gray #f9fafb

- **Icons:**
All icons will be sourced from **Lucide React**, ensuring clarity and consistency.

6.1.3 Layout and Resolution Constraints

- The platform must support responsive layouts:
 - **Minimum mobile resolution:** 360×640 px
 - **Recommended desktop resolution:** 1280×720 px and above
- The layout must auto-adjust for mobile, tablet, and desktop without loss of functionality.
- Tabbing between input fields must follow a **logical and intuitive order** (e.g., name → email → password).

6.1.4 Standard Navigation and Controls

Each screen shall include a consistent **global header** and **navigation bar**, including:

- **Logo:** Clickable logo returns to Dashboard or Home
- **Main navigation menu:**
 - Client: Dashboard, Post Job, Messages, Wallet, Settings
 - Freelancer: Dashboard, Find Work, Proposals, Wallet, Settings
- **User Profile Dropdown:** Profile, Notifications, Settings, Help, Logout
- **Language Toggle:** English GB / Urdu PK
- **Support Button:** Available on all pages, redirects to Help/FAQ section

6.1.5 Shortcut Keys

The following keyboard shortcuts shall be supported on desktop browsers:

- Ctrl + P: Post a Job
- Ctrl + M: Open Messages
- Ctrl + D: Go to Dashboard
- Esc: Cancel or close modals

6.1.6 Message Display Conventions

- **Success Messages:**
 - Displayed in a green banner with a checkmark icon.
 - Automatically dismiss after 5 seconds.
- **Error Messages:**
 - Displayed in a red banner with an exclamation or error icon.
 - Requires manual dismissal by the user (e.g., via a close button).
- **Warning Messages:**
 - Displayed in an orange banner with a warning triangle icon.
 - Automatically dismiss after 8 seconds.

- **Informational Messages:**

- Displayed in a blue banner with an information icon.
- Automatically dismiss after 6 seconds.

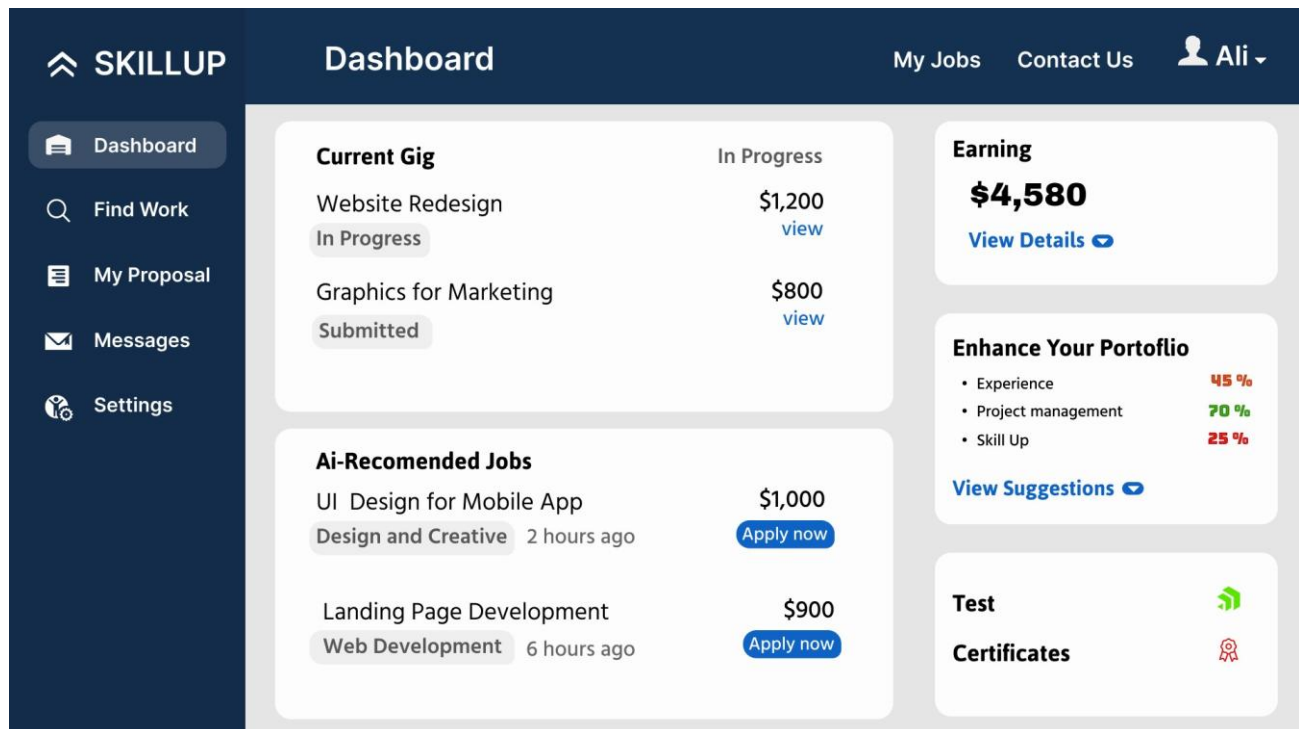
Messages shall be shown as **modals or lightboxes** for critical actions such as submitting proposals or releasing payments.

6.1.7 Localization and Accessibility

- The interface shall be **bilingual** (English and Urdu), selectable via a toggle in the header.
- All RTL (Right-to-Left) support for Urdu must be tested for layout accuracy.
- The UI will include:
 - High contrast color modes
 - Tooltips on icons
 - ARIA labels and keyboard navigability

6.1.8 Visual Mock-ups

Freelancer



SKILLUP

Dashboard

Find Work

My Proposal

Messages

Settings

Dashboard

My Jobs

Contact Us

Ali

Graphic Designer Needed

Category: Design

Budget: \$ 300

Deadline: 2 Weeks

View Detail and bid

E-Commerce Site

Category: Web Development

Budget: \$ 500

Deadline: 4 Weeks

View Detail and bid

1 2 3 Next

Category

key word

key word

key word

Matching Jobs

Web Development

View Details

Redesign Site

View Details

SKILLUP

Dashboard

Find Work

My Proposal

Messages

Settings

Dashboard

My Jobs

Contact Us

Ali

Submit Proposals

Mobile App UI/UX Designer

pending

Bid: PKR 15,000 | TimeLine: 5 Days

Your Message: i'm confident I can design clean, User-friendly mobile UI...

View Job

Edit

Withdraw

Mobile App UI/UX Designer

Hired

Bid: PKR 15,000 | TimeLine: 5 Days

Your Message: i'm confident I can design clean, User-friendly mobile UI...

View Job

Message Client

Client:

SKILLUP

Dashboard

Post a Job

Proposal

Messages

Settings

Dashboard

4 JOBS POSTED

18 PROPOSALS RECEIVED

3 ONGOING PROJECTS

\$ 980 TOTAL SPENT

Recent jobs

Title	Posted on	Status	Proposal	Action
UI/UX Design for Mobile App	March 25 , 2025	Active	6	View
E-Commerce Website	March 20 , 2025	Completed	12	View
SEO of Blog Site	March 18 , 2025	Active	9	View

SKILLUP

Dashboard

Post a Job

Proposal

Messages

Settings

Dashboard

My Jobs

Contact Us

Ali

Job Title

e.g . Build a portfolio website

Job Description

Describe your job requirements....

Category

-- Select Category--

Required Skills

e.g. HTML ,CSS , React

Budget Type

Fixed

Hourly

Budget Amount

e.g. 1500

Deadline

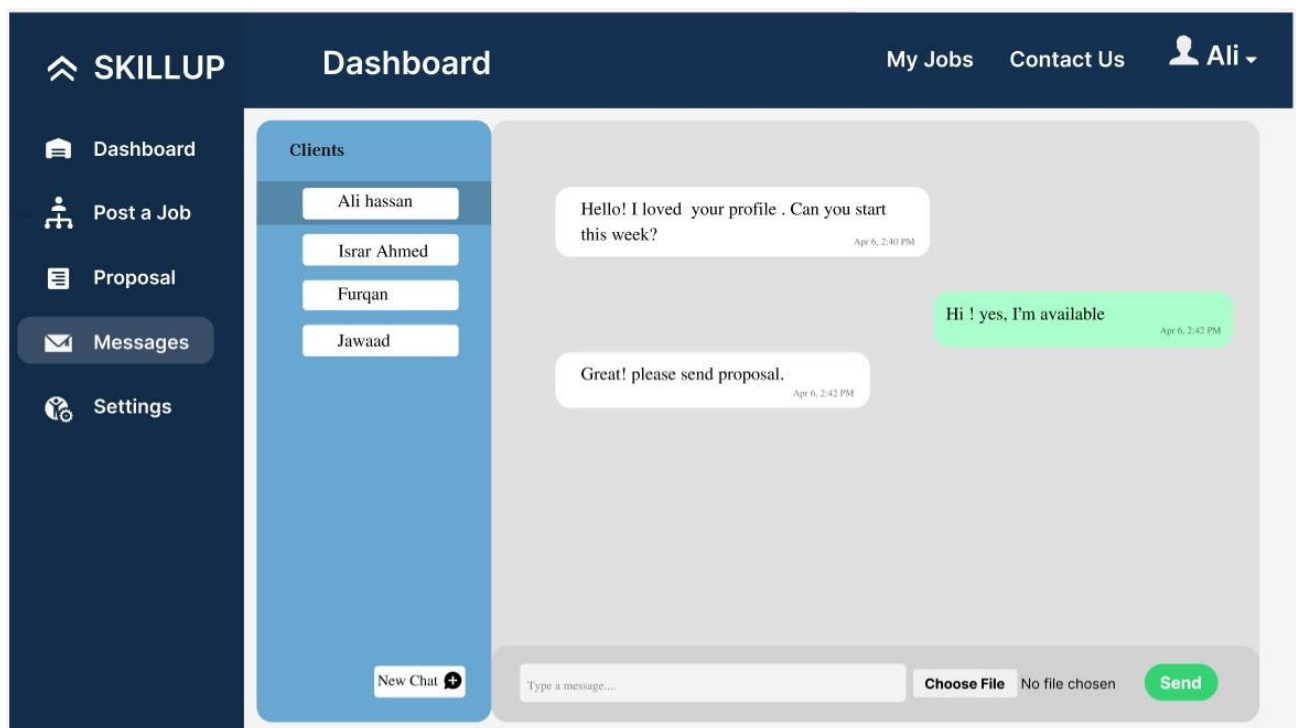
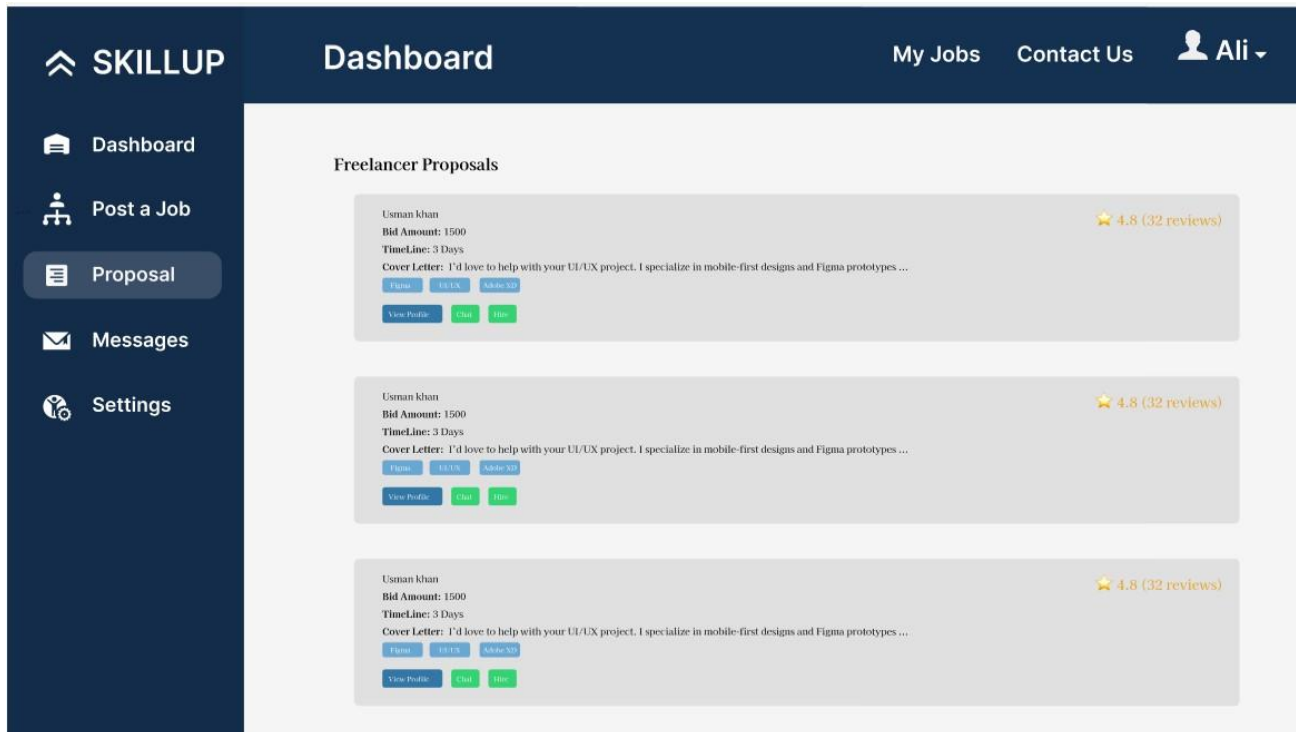
mm / dd / yyyy

Attach File (optional)

Choose File

No file chosen

Post Job



6.2 Software Interfaces

SI-1: MongoDB Atlas

SI-1.1: The SkillUp platform shall connect to MongoDB Atlas as its primary cloud database for storing and retrieving user data, project details, messages, and transaction records.

SI-2: Real-Time WebSocket Communication

SI-2.1: The system shall use WebSocket (via Socket.IO) for bi-directional, real-time communication between users and the server.

SI-3: Payment APIs (JazzCash, Easypaisa, Stripe)

SI-3.1: The platform shall connect with JazzCash, Easypaisa, and Stripe APIs for processing user wallet top-ups and escrow-based transactions.

SI-4: GitHub Webhooks (CI/CD Integration)

SI-4.1: The admin dashboard or developer environment shall integrate with GitHub Webhooks to automatically trigger backend deployments when new code is pushed to the production branch.

SI-5: Express.js RESTful API Layer

SI-5.1: The backend shall expose RESTful APIs to the frontend via Express.js using token-based authentication (JWT).

SI-6: AI Integration (OpenAI)

SI-6.1: The platform shall integrate with Open AI APIs to auto-generate job descriptions and assist freelancers in writing professional proposals.

6.3 Hardware Interfaces

HI-1: The platform is a web application and does not require special hardware interfaces. The platform is entirely cloud-based. However, it shall support:

- Camera and file access from mobile/desktop devices for uploading CNIC or portfolio files
- Print support for downloadable invoices or certificates

6.4 Communication Interfaces

CI-1: The system shall send email notifications (e.g., registration confirmation, project updates) using services Nodemailer.

CI-2: The system shall support SMS notifications for transaction alerts and password resets through local SMS gateway APIs.

CI-3: The frontend shall communicate with the backend using HTTPS REST API calls.

CI-4: The system shall support real-time WebSocket communication for live messaging and status updates.

CI-5: System logs and monitoring data shall be sent to admin consoles via secure internal APIs or webhook integrations.

7. References

Tailwind Labs.

“Tailwind CSS – Utility-First UI Framework.”

Internet: <https://tailwindcss.com/docs>, [Accessed: June. 1, 2025].

Google.

“Material Design Guidelines.”

Internet: <https://m3.material.io/>, [Accessed: June. 1, 2025].

Figma.

“Figma – UI/UX Mockup and Design Tool.”

Internet: <https://www.figma.com/> [Accessed: May. 27, 2025].

H. A. Soares and R. S. Moura,

“A methodology to guide writing Software Requirements Specification document,”

in *Proceedings of the 2015 IEEE Frontiers in Education Conference (FIE)*, El Paso, TX, USA, 2015, pp. 1–8.

Available at: <https://ieeexplore.ieee.org/document/7360001>